Do frontier models expose an attack surface on neural data, and can we evaluate these dangerous capabilities before they can converge into concrete privacy threats?

Motivation: Consumer neurotechnology is generating neural data at increasing scale, and as technological capabilities grow, we face a problem that threatens cognitive freedom and privacy. Legislation has already begun moving ahead of evaluation, with four U.S. states explicitly classifying and protecting neural data as a distinct form of sensitive personal information (Press Release). However, these laws regulate data controllers without regulating AI providers or model behavior, and minimal research has been done on the capability of frontier models to extract sensitive attributes using neural data.

The component capabilities for violating neuro-privacy via inference exist independently. LLMs are shown to be capable of inferring sensitive attributes from text (Staab et al., 2024), images (Tömekçe et al., 2024), and audio (Wang et al., 2025), while neural data raises the stakes by encoding cognitive and clinical information (Xia et al., 2023) that other modalities cannot. Before these components can form a clear pipeline for privacy invasion, we need evaluations that measure model capabilities and alignment with respect to neural data. Similar dangerous capability evaluations already span bio, cyber, and chemistry (e.g., WMDP, CyberSecEval, ChemSafetyBench). By building the first benchmark evaluating frontier models and neural data privacy, we establish a foundation for evaluating dangerous neurocapabilities before they become overwhelmingly apparent.

Experimental Questions: This benchmark will consist of three evaluation scenarios.

Harmful Capability Uplift (Vaccaro et al., 2026)**.** Does a model help an adversary build a neuro-privacy attack pipeline? Harmful capability uplift is specifically measured as the increase in a user's ability to breach privacy given frontier model access relative to the user's harm potential given conventional tools (search engine, literature, etc.).

AI-Alone Attribute Inference. Can the model infer sensitive attributes (e.g., demographic and clinical information) given structured neural data records that don't explicitly contain them? AI-alone attribute inference differs from uplift in that it measures pure AI capability in extracting private information from neural data, given no human expertise.

Over-Refusal. Does the model refuse privacy violations without over-refusing legitimate neuroscience? Matched prompt pairs (Röttger et al., 2023), identical in technical content but differing only in framing, can test whether the model is properly safety-calibrated.

Method and Scope: Text-only, default frontier model behavior is the primary benchmark, with possible extensions in adversarial robustness. Neural data records are structured text derived from neural data (qEEG spectral reports, BCI session logs, participant-level summary statistics, and neuroimaging analysis outputs), avoiding tool use. Scoring will use LLM-judge infrastructure for the uplift and over-refusal evaluations, and top-k accuracy against ground-truth labels for attribute inference.

Future Work: Agentic evaluation (decoder + LLM + tool access), jailbreak robustness, neuroimaging modalities, and extension to adjacent neural-AI threat domains such as neurosecurity.